Chapter 12. Classification of Surfaces

Fundamental groups of surfaces, homology of surfaces, cutting and pasting, the classification theorem

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A **surface** is a 2-manifold. We show that every compact connected surface is homeomorphic to exactly one of: the sphere S^2, a connected sum of n tori T_n, or a connected sum of m projective planes P_m.

12.1 Fundamental groups of surfaces

A **labelling scheme** for a polygonal region (with an even number of edges) assigns to each edge a label and an orientation; identifying edges with the same label according to orientation gives a quotient space, which is a compact surface when each label occurs exactly twice. Schemes are written as words such as $aba^{-1}b^{-1}$.

Example 12.1

aa^{-1} gives S^2; $aba^{-1}b^{-1}$ the torus; aa the projective plane; $abab^{-1}$ the Klein bottle. The n-fold torus T_n comes from $a_1b_1a_1^{-1}b_1^{-1}\cdots a_nb_na_n^{-1}b_n^{-1}$, and the m-fold projective plane P_m from $a_1a_1a_2a_2\cdots a_ma_m$.

Theorem 12.1

If X is obtained from a polygonal region whose scheme identifies all vertices to one point, then X is a wedge of circles (one per label) with a 2-cell attached along the scheme word, so

$$\pi_1(T_n)\cong\langle a_1,b_1,\dots,a_n,b_n \mid [a_1,b_1]\cdots[a_n,b_n]\rangle,\qquad \pi_1(P_m)\cong\langle a_1,\dots,a_m \mid a_1^2\cdots a_m^2\rangle.$$

Proof. The boundary of the polygon maps onto the wedge of circles, the interior is an open 2-cell, and Theorem 11.5 applies. ∎

12.2 Homology of surfaces

The **first homology group** is the abelianization $H_1(X)=\pi_1(X)/[\pi_1(X),\pi_1(X)]$ (for path connected X; independent of the base point).

Theorem 12.2

$H_1(T_n)\cong\mathbb{Z}^{2n}$ and $H_1(P_m)\cong\mathbb{Z}^{m-1}\oplus\mathbb{Z}/2$. Consequently the surfaces $S^2,T_1,T_2,\dots,P_1,P_2,\dots$ are pairwise non-homeomorphic (indeed have non-isomorphic fundamental groups).

Proof. Abelianizing a presentation adds commutation relations. For T_n the relator becomes trivial, giving $\mathbb{Z}^{2n}$. For P_m the relation is $2(a_1+\cdots+a_m)=0$; changing basis to $a_1,\dots,a_{m-1},a_1+\cdots+a_m$ gives $\mathbb{Z}^{m-1}\oplus\mathbb{Z}/2$. ∎

12.3 Cutting and pasting

Two labelling schemes are **equivalent** if they give homeomorphic spaces. The following elementary operations preserve the homeomorphism type: (i) cutting a polygon along a diagonal into two polygons with a new label c on both sides, and its inverse (pasting); (ii) relabelling, and reversing the orientation of all occurrences of a label; (iii) cyclic permutation and inversion of the word; (iv) cancelling an adjacent pair aa^{-1} when the scheme has more than four letters, and its inverse.

Lemma 12.3

The **connected sum** $X\#Y$ (remove an open disc from each and glue along the boundary circles) has scheme w_1w_2 if X, Y have schemes w_1, w_2. Moreover $P_1\#T_1 \cong P_3$, i.e. $aabcb^{-1}c^{-1} \sim aabbcc$.

Idea. The second statement is a sequence of cuts and pastes that moves the pair aa past the commutator, turning $bcb^{-1}c^{-1}$ into two squares; it reflects that a Möbius band absorbs a handle into two cross-caps. ■

12.4 The classification theorem

Theorem 12.4 (Classification of surfaces, polygonal form)

Every space obtained from a polygonal region by a labelling scheme in which each label appears exactly twice is homeomorphic to S^2, T_n or P_m.

Idea. Using elementary operations: (1) cancel pairs aa^{-1} until either the scheme is aa^{-1} (sphere) or none remain; (2) reduce to a scheme in which all vertices are identified to one point (repeatedly cut off a triangle at a vertex of a class not equal to the most frequent one); (3) bring each *twisted pair* $\cdots a \cdots a \cdots$ together as aa; (4) any remaining *untwisted* pairs come in linked couples that can be grouped as commutators $aba^{-1}b^{-1}$; (5) if both aa and commutators appear, use $P_1\#T_1 \cong P_3$ to convert each commutator into two squares. The result is a normal form of type T_n or P_m. ■

Theorem 12.5

Every compact connected surface is homeomorphic to a quotient of a polygonal region as above (Radó: every compact surface can be triangulated), hence to exactly one of S^2, T_n $(n \geq 1)$, P_m $(m \geq 1)$.

The **Euler characteristic** $\chi = V - E + F$ of a triangulation is a topological invariant; $\chi(S^2) = 2$, $\chi(T_n) = 2 - 2n$, $\chi(P_m) = 2 - m$. Orientability distinguishes T_n from P_m, and χ then determines the surface. The Klein bottle is P_2.

12.5 Constructing compact surfaces

Compact surfaces with boundary are obtained by removing finitely many open discs; they are classified by orientability, Euler characteristic and the number of boundary circles. Every compact surface also arises from a polygon, from gluing triangles, or as a branched or unbranched cover of another surface; for instance a k-sheeted covering of T_n is $T_{k(n-1)+1}$ (the Euler characteristic is multiplied by k), and the orientation double cover of P_m is T_{m-1}.